YCCify - Yeshwantrao Chavan College of Engineering

IT701 - Machine Learning | Dr. S. S. Khedkar

Unit 2 - Data Preprocessing & Feature Engineering

Topic: Feature Engineering & Normalization

2.1 Handling Real-World Datasets

Data cleaning is a critical first step in model construction.

Cleaning Techniques:

* Mean/Median/Mode Imputation for Missing Values
* Outlier Detection via IQR and Z-Scores
* MinMax Scaling & Z-Score Standardization

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2.2 Dimensionality Reduction

High dimensional data leads to curse of dimensionality.

Techniques:

* Principal Component Analysis PCA
* Singular Value Decomposition SVD
* Linear Discriminant Analysis LDA